Trae AI Prompt: Optom Capital — Loan CRM

You are building a full-stack Loan CRM web application called "Optom Capital CRM" from scratch.
This is an enterprise-grade internal tool for a loan DSA (Direct Selling Agent) / lending intermediary business. Read every instruction carefully before writing a single line of code.

TECH STACK

- Frontend: React 18 + JavaScript (no TypeScript)
- Styling: Tailwind CSS
- Backend: Supabase (PostgreSQL database + Auth + Storage)
- File Storage: Cloudflare R2 (via presigned URLs or direct upload)
- Routing: React Router v6
- State: React Context + useState/useEffect (no Redux)
- Forms: React Hook Form
- Tables: TanStack Table v8
- Notifications: react-hot-toast
- Icons: Lucide React
- Charts: Recharts
- Date handling: date-fns

AUTH

Single-user app. Use Supabase Auth with email + password only.
- Login page is the entry point if user is not authenticated
- On login, store session, redirect to /dashboard
- Simple logout button in sidebar
- No signup page — user is pre-created in Supabase Auth dashboard
- Protect all routes with a PrivateRoute wrapper
- Persist session across refresh using Supabase's onAuthStateChange

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UI/UX DESIGN SYSTEM

Design aesthetic: Notion-inspired clean layout + Retool enterprise component density.

Rules:
- Background: #F7F7F5 (warm off-white, like Notion)
- Sidebar: #FFFFFF with a subtle right border #E5E5E3
- Cards/panels: #FFFFFF with border #E9E9E7, border-radius 8px, box-shadow: 0 1px 3px rgba(0,0,0,0.06)
- Primary accent: #2F6FED (enterprise blue)
- Danger: #E5484D
- Success: #30A46C
- Warning: #F76B15
- Text primary: #1A1A1A
- Text secondary: #6F6F6F
- Text muted: #AFAFAF
- Font: "Inter" from Google Fonts — weights 400, 500, 600, 700
- Base font size: 13px for table rows/labels, 14px for body, 16px for headings
- All inputs: border #D9D9D6, focus border #2F6FED, border-radius 6px, height 34px
- Buttons: primary (blue fill), secondary (white with border), danger (red fill)
- Table rows: hover background #F5F5F3, row height 40px, divider #EFEFED
- Sidebar width: 220px, fixed, not collapsible
- Top header bar: 52px tall, white, border-bottom #E5E5E3

Component patterns to implement:
- Badge/pill components for statuses (colored dot + text)
- Inline editable fields where appropriate
- Sticky table headers
- Column sort indicators
- Pagination component (10/25/50 rows per page)
- Breadcrumb on all inner pages
- Section dividers with gray headings inside forms
- Confirm dialog before all deletes
- Drawer/slideover panel for creating and editing records (not a new page)
- Empty state with icon + message when no records
- Loading skeleton for all tables and cards

DATABASE SCHEMA (Supabase)

Create the following tables. Use UUIDs as primary keys everywhere. Add created_at and updated_at timestamps to every table with default now() and trigger for auto-update.

Enable Row Level Security (RLS) on all tables. Add a permissive policy for authenticated users to SELECT, INSERT, UPDATE, DELETE.

Table: customers
- id UUID PK
- customer_code TEXT UNIQUE (auto-generated on insert, format: CUS-0001, CUS-0002...)
- name TEXT NOT NULL
- entity_type TEXT CHECK IN ('Individual','Proprietorship','LLP','Pvt Ltd','Partnership')
- pan_number TEXT
- aadhaar_number TEXT
- cin TEXT
- date_of_birth DATE
- incorporation_date DATE
- nationality TEXT DEFAULT 'Indian'
- primary_mobile TEXT
- alternate_mobile TEXT
- email TEXT
- office_phone TEXT
- pa_contact TEXT
- residential_address TEXT
- office_address TEXT
- city TEXT
- state TEXT
- pin_code TEXT
- country TEXT DEFAULT 'India'
- profession_type TEXT CHECK IN ('Doctor','CA','Business Owner','Salaried','Self Employed','Other')
- industry_category TEXT
- business_name TEXT
- business_vintage_years NUMERIC
- employer_name TEXT
- designation TEXT
- company_type TEXT CHECK IN ('MNC','SME','Government','Startup','Other')
- num_employees INTEGER
- gst_number TEXT
- annual_turnover NUMERIC
- net_profit NUMERIC
- monthly_income NUMERIC
- net_worth NUMERIC
- cibil_score INTEGER
- experian_score INTEGER
- equifax_score INTEGER
- crif_score INTEGER
- credit_card_outstanding NUMERIC
- existing_emi_outflow NUMERIC
- total_loan_exposure NUMERIC

- dpd_history TEXT
- writeoff_settlement_history TEXT
- primary_bank TEXT
- current_account_bank TEXT
- savings_bank TEXT
- od_cc_bank TEXT
- banking_vintage_years NUMERIC
- avg_monthly_balance NUMERIC
- monthly_banking_turnover NUMERIC
- cheque_bounce_history TEXT
- last_interaction_date DATE
- next_followup_date DATE
- relationship_notes TEXT
- balance_transfer_opportunity BOOLEAN DEFAULT false
- topup_loan_eligibility BOOLEAN DEFAULT false
- insurance_opportunity BOOLEAN DEFAULT false
- credit_card_offer BOOLEAN DEFAULT false
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

Table: leads
- id UUID PK
- lead_code TEXT UNIQUE (auto-generated: LEAD-0001, LEAD-0002...)
- customer_id UUID FK → customers.id ON DELETE CASCADE
- loan_type TEXT CHECK IN ('Personal Loan','Business Loan','Loan Against Property','Working Capital','Overdraft / CC','Equipment Finance')
- loan_purpose TEXT
- required_amount NUMERIC
- preferred_tenure_months INTEGER
- preferred_bank TEXT
- urgency_level TEXT CHECK IN ('High','Medium','Low') DEFAULT 'Medium'
- lead_source TEXT
- referral_partner_name TEXT
- sales_manager_name TEXT
- sales_manager_contact TEXT
- branch_location TEXT
- lead_creation_date DATE DEFAULT now()
- stage TEXT CHECK IN ('Lead Generated','Qualification Done','Documents Collected','Login to Bank','Underwriting','Sanction Approved','Disbursement Processing','Disbursed','Declined') DEFAULT 'Lead Generated'
- credit_observations TEXT
- structuring_notes TEXT
- risk_flags TEXT
- rejection_reason TEXT

- case_history TEXT
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

Table: collateral
- id UUID PK
- lead_id UUID FK → leads.id ON DELETE CASCADE
- customer_id UUID FK → customers.id ON DELETE CASCADE
- property_type TEXT CHECK IN ('Residential','Commercial','Industrial')
- property_address TEXT
- market_value NUMERIC
- distress_value NUMERIC
- ltv_percentage NUMERIC
- property_ownership TEXT
- property_age_years INTEGER
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

Table: bank_applications
- id UUID PK
- lead_id UUID FK → leads.id ON DELETE CASCADE
- bank_name TEXT NOT NULL
- bank_rm_name TEXT
- rm_contact TEXT
- login_date DATE
- case_id TEXT
- underwriting_status TEXT CHECK IN ('Pending','In Review','Query Raised','Approved','Declined')
- credit_queries TEXT
- sanction_amount NUMERIC
- interest_rate NUMERIC
- loan_tenure_months INTEGER
- emi_amount NUMERIC
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

Table: documents
- id UUID PK
- customer_id UUID FK → customers.id ON DELETE CASCADE
- lead_id UUID FK → leads.id ON DELETE SET NULL
- document_type TEXT CHECK IN ('PAN','Aadhaar','Passport/DL','ITR - Year 1','ITR - Year 2','ITR - Year 3','Balance Sheet','GST Returns','Bank Statement 12M','Title Deed','Sale Agreement','Valuation Report','Other')
- file_name TEXT

- file_url TEXT (R2 public or presigned URL)
- r2_key TEXT (raw R2 object key for deletion)
- status TEXT CHECK IN ('Pending','Received','Verified') DEFAULT 'Pending'
- notes TEXT
- uploaded_at TIMESTAMPTZ DEFAULT now()
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

Table: disbursements
- id UUID PK
- lead_id UUID FK → leads.id ON DELETE CASCADE
- bank_application_id UUID FK → bank_applications.id ON DELETE SET NULL
- sanction_amount NUMERIC
- final_disbursed_amount NUMERIC
- disbursement_date DATE
- bank_payout_percentage NUMERIC
- gross_revenue NUMERIC
- partner_share NUMERIC
- net_revenue NUMERIC
- invoice_raised BOOLEAN DEFAULT false
- payout_received BOOLEAN DEFAULT false
- payment_pending BOOLEAN DEFAULT true
- created_at TIMESTAMPTZ DEFAULT now()
- updated_at TIMESTAMPTZ DEFAULT now()

CLOUDFLARE R2 STORAGE

For document uploads, use Cloudflare R2 via presigned PUT URLs.

Create a Supabase Edge Function (or serverless function) called `get-r2-presigned-url` that:
- Accepts: filename, content_type, folder (e.g. "kyc", "financial", "property")
- Returns: a presigned PUT URL and the final public URL / object key
- Uses R2 credentials from environment variables: R2_ACCOUNT_ID, R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_BUCKET_NAME, R2_PUBLIC_DOMAIN

On the frontend, upload flow:
1. Call the edge function to get presigned URL
2. PUT the file directly to R2
3. Save the returned public URL and r2_key to the documents table in Supabase

For deletion: call another edge function `delete-r2-object` passing the r2_key.

File size limit: 25MB per file.
Allowed types: PDF, JPG, PNG, XLSX, DOCX.
Show upload progress bar during upload.

APPLICATION STRUCTURE

Folder Structure

src/ components/ layout/ Sidebar.jsx Header.jsx Layout.jsx ui/ Button.jsx Input.jsx Select.jsx Badge.jsx Modal.jsx Drawer.jsx Table.jsx Pagination.jsx Skeleton.jsx ConfirmDialog.jsx StatCard.jsx EmptyState.jsx FileUpload.jsx Breadcrumb.jsx pages/ Login.jsx Dashboard.jsx Customers/ CustomerList.jsx CustomerDetail.jsx Leads/ LeadList.jsx LeadDetail.jsx LeadPipeline.jsx BankApplications/ BankApplicationList.jsx Documents/ DocumentsPage.jsx Disbursements/ DisbursementList.jsx Reports/ ReportsPage.jsx context/ AuthContext.jsx lib/ supabase.js r2.js utils.js App.jsx main.jsx

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SIDEBAR NAVIGATION

Fixed left sidebar, 220px wide. Logo at top ("Optom Capital" in blue with a small chart icon).

Nav items with Lucide icons:
- Dashboard (LayoutDashboard)
- Customers (Users)
- Leads (FileText)
- Pipeline (Kanban)
- Bank Applications (Building2)
- Documents (FolderOpen)
- Disbursements (IndianRupee)
- Reports (BarChart2)

Active item: blue left border + blue text + light blue background.
Inactive: #6F6F6F text, transparent background, hover: #F5F5F3.

PAGE SPECIFICATIONS

1. LOGIN PAGE
- Centered card on #F7F7F5 background
- "Optom Capital" heading in blue

- "Loan CRM" subtitle in gray
- Email + Password fields
- "Sign In" primary button (full width)
- Show/hide password toggle
- Error message display for wrong credentials
- No signup link

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2. DASHBOARD

Top row — 5 stat cards:
- Total Leads (count from leads table)
- Active Cases (leads not in Disbursed or Declined)
- Sanctioned Cases (stage = 'Sanction Approved')
- Disbursed Cases (stage = 'Disbursed')
- Monthly Revenue (sum of net_revenue from disbursements this month, format as ₹)

Second row — 2 charts side by side:
Left: Bar chart — Monthly Disbursed Amount (last 6 months) using Recharts
Right: Pie chart — Lead Stage Distribution (count by stage) using Recharts

Third row — 2 tables side by side:
Left: Top Performing Banks (group bank_applications by bank_name, sum sanction_amount, show top 5)
Right: Top Referral Partners (group leads by referral_partner_name, count leads, show top 5)

Fourth row — Recent Leads table (last 10 leads, columns: Lead ID, Customer, Loan Type, Amount, Stage, Date)

All numbers formatted as Indian Rupees: ₹ with Indian number system (lakhs/crores).
All stat cards have a subtle trend indicator.

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3. CUSTOMERS LIST PAGE

URL: /customers

Header: "Customers" heading + "Add Customer" primary button (top right)

Search bar (searches name, PAN, mobile, email).

Filter dropdowns: Entity Type, Profession Type, City.

Table columns (sortable):
- Customer ID
- Name
- Entity Type (badge)
- Mobile
- Email
- City
- CIBIL Score (color coded: green ≥750, yellow 650-749, red <650)
- Next Follow-up (red if overdue)
- Actions (View, Edit, Delete)

Click row or View → /customers/:id
Edit → opens Drawer with full form
Delete → ConfirmDialog → soft delete or hard delete

Pagination: 25 per page default.

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4. CUSTOMER DETAIL PAGE

URL: /customers/:id

Breadcrumb: Customers > [Customer Name]

Left column (60%): Tabbed sections
- Tab 1: Basic Info (all identity + contact + address fields)
- Tab 2: Business Profile (profession, business, GST, income fields)
- Tab 3: Credit Profile (all credit scores + bureau data, outstanding, DPD)
- Tab 4: Banking Info (bank relationships, balance, turnover, bounce history)
- Tab 5: Cross-Sell (4 checkboxes + relationship notes, follow-up date)

Right column (40%):
- Top: Mini stats (Total Leads, Active Leads, Total Disbursed)
- Leads associated with this customer (mini table: Lead ID, Loan Type, Amount, Stage)
  "Add Lead" button to create a new lead linked to this customer
- Documents panel: list of all docs for this customer, upload button

Edit button (top right of each tab section) opens inline editing mode for that section.
Save / Cancel buttons appear when in edit mode.

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5. LEADS LIST PAGE

URL: /leads

Header: "Leads" + "Add Lead" button

Search: Lead ID, Customer Name, Referral Partner

Filters: Loan Type, Stage, Urgency Level, Sales Manager, Date Range

Table columns:
- Lead ID
- Customer Name (clickable → customer detail)
- Loan Type (badge)
- Required Amount (₹)
- Stage (colored status badge)
- Urgency (badge: red=High, yellow=Medium, gray=Low)
- Sales Manager
- Lead Date
- Actions (View, Edit, Delete)

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6. LEAD DETAIL PAGE

URL: /leads/:id

Breadcrumb: Leads > [Lead ID]

Top section: Lead header card
- Lead ID, Customer Name (link), Loan Type, Required Amount
- Stage selector (dropdown that updates stage in real-time)
- Urgency badge
- Created date

Tab sections:
- Tab 1: Loan Details (all loan requirement fields)
- Tab 2: Collateral (collateral details form — show/hide based on loan type: show for LAP)
- Tab 3: Bank Applications
  - Table of all bank_applications for this lead
  - Columns: Bank, RM Name, Login Date, Case ID, UW Status, Sanction Amount, Rate, Tenure, EMI
  - "Add Bank Application" button → inline drawer form
  - Edit/Delete per row

- Tab 4: Disbursement
  - Show disbursement record if exists, else "Record Disbursement" button
  - All disbursement fields in a form card
  - Invoice Raised / Payout Received toggles as toggle switches
- Tab 5: Documents
  - Filter by document_type
  - Table: Doc Type, File Name, Status, Uploaded Date, Actions (View, Download, Delete)
  - Upload button: opens file picker, uploads to R2, saves to documents table
  - Status badge: Pending (gray), Received (blue), Verified (green)
- Tab 6: Risk & Notes
  - Text areas for: Credit Observations, Structuring Notes, Risk Flags, Rejection Reason, Case History

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7. PIPELINE PAGE (KANBAN VIEW)

URL: /pipeline

A kanban board with 9 columns — one per deal stage:
1. Lead Generated
2. Qualification Done
3. Documents Collected
4. Login to Bank
5. Underwriting
6. Sanction Approved
7. Disbursement Processing
8. Disbursed
9. Declined

Each card shows:
- Lead ID
- Customer Name
- Loan Type
- Required Amount (₹)
- Urgency badge

Cards are draggable between columns (use @hello-pangea/dnd or react-beautiful-dnd).
On drop, update the lead's stage in Supabase.
Each column header shows count of cards.
Columns scroll vertically independently.
Board scrolls horizontally.
Filter by: Loan Type, Sales Manager, Urgency

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8. BANK APPLICATIONS PAGE

URL: /bank-applications

Header: "Bank Applications" + "Add Application" button

Search: Customer name, bank name, case ID

Filters: Bank Name, Underwriting Status, Date Range

Table columns:
- Application ID
- Lead ID (link → lead detail)
- Customer Name
- Bank Name
- RM Name
- Login Date
- Case ID
- UW Status (badge)
- Sanction Amount (₹)
- Interest Rate (%)
- Tenure (months)
- EMI (₹)
- Actions

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9. DOCUMENTS PAGE

URL: /documents

Global document repository.

Search: file name, customer name.

Filters: Document Type, Status, Customer.

Table columns:
- Document Type
- Customer Name
- Lead ID
- File Name (clickable — opens in new tab)

- Status (badge)
- Uploaded At
- Actions (View, Download, Delete)

Upload button at top: select customer, lead (optional), doc type, then upload file.

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10. DISBURSEMENTS PAGE

URL: /disbursements

Header: "Disbursements"

Summary row at top (4 mini cards):
- Total Disbursed (sum final_disbursed_amount)
- Total Gross Revenue (₹)
- Total Net Revenue (₹)
- Pending Payouts (count where payment_pending = true)

Table columns:
- Lead ID
- Customer Name
- Bank
- Disbursed Amount (₹)
- Date
- Payout %
- Gross Revenue (₹)
- Partner Share (₹)
- Net Revenue (₹)
- Invoice Raised (✓/✗)
- Payout Received (✓/✗)
- Payment Pending (badge)
- Actions (Edit)

Filter: Date Range, Bank, Payout Received status.

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11. REPORTS PAGE

URL: /reports

Section 1: Revenue Report

- Date range picker (month/quarter/custom)
- Total leads, total disbursed amount, gross revenue, net revenue in that period
- Bar chart: Net Revenue by Month

Section 2: Lead Funnel
- Stage-by-stage breakdown (count + %)
- Horizontal bar chart

Section 3: Bank Performance
- Table: Bank Name | Applications | Sanctioned | Declined | Total Sanctioned Amount | Avg Interest Rate

Section 4: Top Referral Partners
- Table: Partner Name | Leads Sent | Disbursed | Conversion % | Total Revenue

Section 5: Collection Status
- Table of disbursements: Invoice Raised / Payout Received / Pending breakdown

Export to CSV button on every table.

KEY UTILITY FUNCTIONS (lib/utils.js)

- formatINR(amount) → formats number as ₹1,23,45,678 (Indian format)
- generateCode(prefix, count) → e.g. CUS-0001, LEAD-0042
- formatDate(date) → DD MMM YYYY
- getStageColor(stage) → returns Tailwind color classes per stage
- getScoreColor(score) → green/yellow/red based on CIBIL range
- truncateText(text, length)
- isOverdue(date) → returns true if date is past today

SUPABASE SETUP

In lib/supabase.js:
- Initialize Supabase client using VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY from .env
- Export single instance

Auth helpers:
- signIn(email, password)
- signOut()

- getCurrentUser()
- onAuthStateChange(callback)

Data helpers (use these patterns consistently):
- Always use .order('created_at', { ascending: false }) on list queries
- Always handle error + data from Supabase responses
- Use toast.error() on all errors, toast.success() on all mutations

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FORM BEHAVIOR

For all Create/Edit Drawers:
- Use React Hook Form for all forms
- Validate required fields before submit
- Show inline field-level error messages in red below each input
- Disable submit button while submitting, show spinner
- On success: close drawer, refresh table, show success toast
- On error: keep drawer open, show error toast

Drawer:
- Slides in from the right
- Width: 600px on desktop
- Has a header with title and X close button
- Scrollable content area
- Sticky footer with Cancel + Submit buttons
- Dark overlay on main content (not full black, use rgba)

GLOBAL BEHAVIORS

- All monetary values displayed in Indian Rupee format with ₹ prefix
- All dates displayed as "12 Jan 2025" format
- All tables have a "No records found" empty state with an icon
- All delete actions require a confirm dialog with the record name
- Loading states: skeleton loaders on initial page load, spinner on actions
- All pages have a breadcrumb
- Toasts appear top-right, auto-dismiss in 3s
- Responsive down to 1280px width minimum (not mobile-first)
- Browser tab title: "Optom Capital CRM"

ENVIRONMENT VARIABLES (.env)

VITE_SUPABASE_URL=
VITE_SUPABASE_ANON_KEY=
VITE_R2_WORKER_URL= (the Cloudflare Worker or edge function endpoint for presigned URLs)

DELIVERABLES

Build the complete application with all pages, components, and database integration as specified above. Do not skip any page or section. Do not use placeholder data — all data must come from Supabase in real-time.

Start with:
1. Supabase schema SQL (all CREATE TABLE statements with indexes and RLS policies)
2. Project scaffolding (vite + react setup)
3. Design system components (ui/ folder)
4. Auth flow
5. Layout (sidebar + header)
6. Dashboard
7. Then all remaining pages in order

Write production-quality code. Add comments only where the logic is non-obvious.
Handle all loading, error, and empty states on every page.

Before pasting into Trae AI:

- Have your Supabase project created and grab the URL + Anon Key
- Have your R2 bucket set up and a Cloudflare Worker ready for presigned URLs (or ask Trae to scaffold the Worker too — just append that ask)
- If you want Trae to also build the Supabase Edge Function for R2 presigned URLs, add that explicitly at the end

Chunking tip: If Trae AI has a context limit, break it into 3 passes — (1) DB schema + auth + layout, (2) Customers + Leads + Pipeline, (3) Bank Applications + Documents + Disbursements + Reports.